25901919 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/MGU-BBA (Honours)/MGU-BCA (Honours) Special Examinations, July 2026 (for 2024 Admission Regular Examinations, October 2025)

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY202 - INTRODUCTION TO SPACE PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. What is cold plasma? [U] / [CO1]
2. Mention two challenges faced in human spaceflight missions. [Ap] / [CO6]
3. Why is studying coronal mass ejections important for Earth's technology systems? [U] / [CO4]
4. What role does space weather forecasting play in power grid protection ? [U] / [CO5]
5. How do solar energetic particles affect satellites ? [U] / [CO2]
6. What is space weather ? [U] / [CO2]
7. What is the main cause of solar flares? [K] / [CO2]
8. What is the main process behind the energy production in the Sun's core? [K] / [CO4]
9. State any two applications of the measurement of TEC from GPS data. [K] / [CO5]

10. Briefly discuss the magnetic reconnection in the magnetosphere of Jupiter. [K] / [CO3]

11. Define sporadic E. [K] / [CO3]

12. State the relationship of plasma frequency to electron density. [K] / [CO3]

13. Is exosphere isothermal? Justify your answer with reason. [U] / [CO4]

14. Why is a southward IMF orientation important for magnetospheric activity? [U] / [CO5]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. Explain the process of geomagnetic storms caused by solar activity and their impact on Earth. [U] / [CO1]

16. Describe the growth of space physics from early aurora observations to the discovery of Earth's Van Allen belts. [An] / [CO6]

17. With a neat diagram explain the atmosphere of Sun. [U] / [CO2]

18. Explain how SOHO's location at L1 benefits continuous solar observation. [U] / [CO4]

19. Explain the origin of the solar wind and how its properties vary with solar activity. [U] / [CO5]

20. With a neat diagram, describe the structure of Earth's magnetosphere. [U] / [CO3]

21. Discuss the primary and secondary processes which contribute to the production of charge carriers in ionosphere. [U] / [CO4]

22. Describe the formation of a bow shock in front of a planetary magnetosphere. [U] / [CO5]

[5x6 = 30]